Product use and verification guide

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Repository

https://github.com/yerkebttt/municipal-air-quality-batch-ingestion

1. Clean start

```
docker compose down -v
docker compose up --build
```

The first command removes the database volume for this project. The second command starts MongoDB, waits for a successful health check, downloads the official nested UCI archive and loads the data in repeatable batches.

2. Inspect health and data quality

```
docker compose run --rm ingest python -m src.monitoring
docker compose run --rm ingest python -m src.queries quality
```

The health report should show a completed run and counters for source, accepted, warning, rejected, inserted and existing records. The verified run processed and stored all 420,768 rows. It reported 383,964 accepted observations, 36,804 warnings and zero structural rejection.

3. Exercise user contracts and latency gate

```
docker compose run --rm ingest python -m src.queries trend --station Aotizhongxin --pollutant pm25
docker compose run --rm ingest python -m src.queries compare --pollutant pm25
docker compose --profile verification up --build --exit-code-from verify
```

The benchmark checks recent readings, a daily PM2.5 trend, a comparison of 12 stations and a quality summary. All queries passed the 2,000 ms target. Their measured times were 15.931 ms, 16.505 ms, 21.217 ms and 221.352 ms. The command records results in query_benchmarks and exits with an error code if a future query misses the target. After a repeated load, the measurement count must not increase.

Operational boundary

The threshold configuration is an example and creates cases for review only. The prototype cannot notify citizens. Before production use, the historical fixture must be replaced by an agreed municipal feed. The municipality also needs verified station metadata, approved policy thresholds, access control, TLS, backups, replication and testing with concurrent users.